Heterogeneous Enforcement Within a Common Tax System
Tax Office Aggressiveness and Firm Outcomes*

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Abstract

We study differences in enforcement behavior across tax offices and their effects on firms. Using administrative data on the universe of tax litigation involving Italian corporations, we construct a novel measure of enforcement behavior: the propensity to appeal judgments in favor of firms, adjusting for case, firm, and court characteristics. Appeals prolong litigation and raise costs for firms, and reveal how aggressively offices pursue contested tax claims. We document substantial within-province variation: moving from the least to the most aggressive office raises the probability of appeal by 18.7 percentage points on average. More aggressive offices may affect firms beyond litigation by increasing administrative burdens and strengthening deterrence against tax evasion. To estimate the effects of office aggressiveness, we exploit a reform that centralized enforcement and changed firms' assigned offices. We find substantial negative effects concentrated among more vulnerable firms and in sectors with greater opportunities for tax evasion. Among smaller and liquidity-constrained firms, a 10-percentage-point increase in aggressiveness reduces revenues by 3–4% and increases cumulative exit probabilities by 0.3–0.9 percentage points. Overall, our findings highlight the real economic consequences of differences in tax enforcement.

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1 Introduction

The economic consequences of a tax system depend not only on the rules it establishes, but also on how those rules are enforced. While tax rates, exemptions, and reporting requirements determine firms' statutory obligations, their implementation involves discretionary decisions by tax authorities, such as which taxpayers to audit, which claims to pursue, and how to resolve disputes. Even within a common tax system, differences in these decisions may expose firms and individual taxpayers to different enforcement burdens. This may be particularly relevant in decentralized tax systems, where common rules are implemented by local tax authorities. Yet, despite the central role of tax administration in the functioning of the economy, we know relatively little about how enforcement practices differ across local tax authorities and whether these differences have real economic consequences.

Two empirical challenges have hindered the study of the heterogeneous implementation of the tax system and its consequences. First, enforcement practices are rarely observed and difficult to measure. Administrative data typically capture at best some of the outcomes of enforcement decisions, rather than the full set of choices available to tax authorities. Moreover, observed outcomes reflect the composition of taxpayers and cases faced by tax authorities, not just differences in enforcement behavior. Second, even if measured, identifying the effects of enforcement practices on economic activity is challenging because firms are not randomly assigned to tax authorities. Differences in enforcement may be correlated with local economic conditions or firm characteristics that independently affect economic outcomes.

This paper studies whether tax offices operating under a common national tax system differ systematically in their enforcement choices and whether these differences have real economic consequences. We use administrative records on the universe of tax litigation cases and balance-sheet data covering the population of Italian limited liability firms. These data allow us to address the two empirical challenges by combining a novel measure of enforcement intensity at the office level with quasi-experimental variation in the assignment of firms to tax offices. Our measure exploits one enforcement margin that we observe in full: whether an office appeals a first-instance judgment that it lost, conditional on case, firm, and court characteristics. We first provide systematic evidence of substantial heterogeneity in enforcement behavior across offices, within the same formal tax system. We then exploit a reform that merged and centralized tax offices to estimate the effects of these differences on firm outcomes. We find that differences in enforcement by local offices have economically meaningful consequences for firms, particularly those that are smaller or financially vulnerable.

The Italian tax administration provides a well-suited setting to study heterogeneity in enforcement practices. The Italian Revenue Agency administers national taxes through a

decentralized organization in which territorial offices carry out operational activities, including assessments, audits, and the management of subsequent litigation. Before a centralization reform implemented between 2009 and 2011, these activities were performed by more than 280 Local Offices, whose jurisdictions consisted of a distinct and relatively small group of municipalities, within a province. Except for very large taxpayers, all firms in a jurisdiction were assigned to the same local tax office, based on the businesses' fiscal residence. Importantly, offices operated under common national tax rules and guidelines, and offices in the same province also faced the same tax courts. This organization allows us to study differences in enforcement behavior across authorities while holding fixed common legal and institutional factors.

Italy also provides a high-stakes setting for studying tax enforcement. Despite its developed tax administration, tax evasion remains a substantial policy challenge. In 2023, the estimated tax gap, the difference between taxes owed under full compliance and actual collections, amounted to more than 100 billion Euros, or 17% of potential tax revenues.[1] Previous research in the Italian context has shown that incentives faced by individual auditors influence audit targeting (Porto et al., 2013), while constraints on local auditing capacity can amplify tax evasion (Galbiati and Zanella, 2012). Additionally, large-scale anti-evasion interventions have generated important fiscal and electoral responses, highlighting also the political salience of tax enforcement (Casaburi and Troiano, 2016).

We first construct a measure of tax-office aggressiveness based on offices' propensity to appeal first-instance judgments in favor of firms. This margin is particularly informative because we observe the complete set of judgments that each office could appeal and its decisions to do so. However, observed appeal rates may reflect differences in the composition of cases and firms faced by each office, rather than systematic differences in enforcement behavior. To isolate the latter, we estimate office fixed effects, flexibly controlling for case and firm characteristics and the judicial environment. Since tax courts operate at the provincial level, controlling for court fixed effects restricts identification to differences in appeal propensities within provinces only. Finally, we apply Empirical Bayes shrinkage to account for differences in the precision of the estimates across offices (Walters, 2024).

We document substantial heterogeneity in tax office aggressiveness, even among tax offices operating within the same province. Among the local offices operating prior to the centralization reform, the difference between the least and the most aggressive office within a province averages 18.7 percentage points. In relative terms, this is about 31% of the average probability of appeal by a tax office, which is 61%. This dispersion is remarkable given

[1]Source: Ministry of Economy and Finance, *Relazione sull'economia non osservata e sull'evasione fiscale e contributiva* (2026), Table I.2 (link).

that offices operate under common national tax rules and guidelines and, within provinces, face the same tax courts. We then provide evidence supporting the interpretation of our measure as a proxy for broader differences in enforcement practices. More aggressive offices are associated with more frequent tax litigation, higher rates of losing the appeals they pursue, and a greater share of staff time devoted to litigation. These correlations are consistent with more aggressive offices being willing to pursue more marginal tax claims and allocating resources accordingly.

The second part of the paper documents that the differences in enforcement practices we measure have large impacts on economic activity. We identify the effects of tax-office aggressiveness on firm outcomes by exploiting a centralization reform that changed the enforcement authority assigned to firms. Within each province, before 2009, firms were assigned to different local offices depending on their municipality of fiscal residence. Each of these local offices was characterized by its own level of aggressiveness, which we measure separately. Following the reform, which was completed in 2011, all firms within a province were assigned to a common provincial office. Thus, the reform induced different changes in the aggressiveness of the assigned tax office depending on firms' original local office.

We combine two sources of variation: cross-sectional variation in the change in aggressiveness generated by the reform, and time variation around the implementation of the reform. We formalize the identification strategy in a potential outcome framework, where treatment is continuous and effects are a function of the change in aggressiveness. We show that our estimator is comparable to a continuous difference-in-differences design, but requires us to impose restrictions on the heterogeneity of treatment effects across firms, because no firm is untreated. That is, no firm remains at the pre-reform level of aggressiveness. Firm fixed effects absorb time-invariant differences, while province-by-year fixed effects account for common local shocks and province-wide consequences of the reform. While the underlying assumptions are stronger than the usual parallel trends assumption, the within-province quasi-experimental variation and the richness of our data allow us to empirically assess their plausibility and relax the stricter assumptions of treatment effect homogeneity.

Increased tax-office aggressiveness reduces firm performance, particularly for more vulnerable firms. On average, we find that revenues and profitability decline. A 10-percentage-point increase in aggressiveness reduces firm revenues by approximately 1.2% and return on assets by 0.1 percentage points, equivalent to about 3.8% of its pre-reform level.[2] Both estimates are statistically significant at the 10% level. By contrast, the average effects on employment and firm survival are small and statistically insignificant. Event-study estimates show no

[2]We use a 10-p.p. change as a simple reference, which is close to the average change in absolute value induced by the reform, equal to 12.6 p.p.

systematic differences in pre-reform trends, supporting the validity of our identification strategy, and indicate that the negative effects on revenues and profitability emerge rapidly after the reform and persist over time. We also find evidence of a gradual increase in the probability of exiting, suggesting that the consequences for firm survival may take longer to materialize.

The magnitude and heterogeneity of the effects on firm performance help us shed light on the economic mechanisms at play. Evidence from the literature suggests that aggressive enforcement may increase the administrative and financial costs of dealing with tax authorities (Blumenthal and Slemrod, 1992; Hanlon et al., 2017; Belnap et al., 2024; Cowx, 2025), strengthen incentives for tax compliance (Slemrod et al., 2001; Kleven et al., 2011; Pomeranz, 2015; Carrillo et al., 2017), and induce a reallocation of economic activity across competing firms (Di Marzio et al., 2026). Consistent with these mechanisms, we find that adverse effects are concentrated among smaller and liquidity-constrained firms, which may be less able to absorb enforcement-related costs, and in service sectors characterized by greater opportunities for tax evasion. Larger firms, by contrast, are generally less affected and, if anything, experience some increases in employment, consistent with potential reallocation of business to larger firms. The magnitudes are economically large: for liquidity-constrained firms, a 10-percentage-point increase in aggressiveness raises cumulative exits by 0.9 percentage points and reduces revenues and employment by 4.0% and 0.8%, respectively. Among smaller firms, the same increase raises cumulative exits by 0.3 percentage points and reduces revenues by 3.2%.

Related literature. Our paper contributes to the literature on how tax systems affect firm outcomes and the allocation of economic activity. A large literature shows that differences in tax rates and statutory rules more generally affect firms' size, investment and location decisions, potentially generating distortions in the allocation of resources (Devereux and Griffith, 1998; Suárez Serrato and Zidar, 2016; Giroud and Rauh, 2019; Fajgelbaum et al., 2019; Akcigit et al., 2021; Zwick, 2021; Chodorow-Reich et al., 2024; Kennedy et al., 2026). Related work examines fiscal competition and the role of state and local tax incentives in attracting firms and employment (Agrawal et al., 2022; Slattery and Zidar, 2020). This literature has largely focused on changes to the tax system and differences in its formal rules. We show that differences in the implementation of rules, within a common tax system, also matter for firm outcomes. By measuring differences in enforcement practices across tax offices and exploiting quasi-exogenous variation in office assignment, we provide micro-level estimates of an additional cause of heterogeneity in the tax environments that firms effectively face. Our findings suggest that heterogeneous enforcement can affect the allocation of economic

activity across firms, even in the absence of differences in statutory tax rules.

Second, we contribute to the literature on tax administration and enforcement. This growing literature is largely based on evidence from developing countries and has focused on how enforcement capacity, tax design, and the organization and incentives of tax administrators affect tax collection (Best et al., 2015; Khan et al., 2016; Basri et al., 2021; Balán et al., 2022; Bergeron et al., 2024; Decet, 2025). Related research documents size-dependent differences in enforcement and their implications for tax compliance (Almunia and Lopez-Rodriguez, 2018; Bachas et al., 2019; Di Gregorio et al., 2024). Unlike these studies, we focus on differences in enforcement arising from the discretionary choices of tax offices operating under common rules and guidelines. Moreover, we focus on firm outcomes beyond compliance, adding evidence on the real effects of tax enforcement. In doing so, we complement research on the effects of tax audits on firm performance (Belnap et al., 2024). We study the broader economic consequences of enforcement practices for all firms assigned to a tax office, regardless of whether they are directly involved in tax audits and litigation.

Finally, our paper relates to the literature on how the functioning of public institutions affects firm performance. This literature generally draws on evidence from judicial institutions and shows that differences in court efficiency and contract enforcement affect firm size, access to credit, and investment (Ponticelli and Alencar, 2016; Giacomelli and Menon, 2017; Ichino et al., 2023). More closely related, Gianfreda and Vallanti (2017) show that delays in labor courts affect job flows and productivity, while Cahuc et al. (2024) document how differences in judges' decisions affect firm employment and survival. We extend this literature by studying a different public institution, which is key for economic activity: the tax authority. Our findings show that differences in behavior of local branches of the national tax authority are large and have sizeable consequences for firm performance, particularly among smaller and financially constrained firms.

2 Institutional Context

2.1 The Italian Revenue Agency

The Italian Revenue Agency (*Agenzia delle Entrate*) was established by Legislative Decree 300/1999 and has operated since January 2001. It is a public body that took over the functions previously performed by the Revenue Department of the Ministry of Finance, and it is supervised by the Ministry of Economy and Finance. Its mandate covers the administration and assessment of national taxes, including personal and corporate income taxes and value-added tax (Agenzia delle Entrate, 2005). The Agency has its own statute and administrative

regulations, adopted by a management board and proposed by the Agency's Director. Its structure is hierarchical: central and regional offices mainly plan, direct, coordinate, and monitor, while territorial offices perform the operational activities.

Like tax administrations in many developed countries, the Agency has a territorially decentralized structure: operational tasks such as tax audits are carried out by offices that each serve a defined geographic area rather than by a single national body. A taxpayer's office is determined by the *domicilio fiscale*, which is the municipality of registered residence for individuals and the municipality of the registered office for firms. A taxpayer therefore cannot choose its office, and its tax matters are handled by the office whose territory includes that municipality.[3] As a result, which office oversees a taxpayer depends only on where the taxpayer is registered.

2.1.1 Territorial Organization Before the Reform

Before the centralization reform, the Agency's territorial organization had four tiers: a national directorate, regional offices, Provincial Offices, and subprovincial Local Offices (*Uffici Locali*). Its 285 Local Offices each covered a distinct set of municipalities within a single province, amounting to approximately 2.7 Local Offices for each of the 106 Provincial Offices. Provincial Offices mainly coordinated the activities of the Local Offices in their territory, whereas Local Offices carried out the Agency's operational tasks. In particular, Local Offices conducted tax audits and represented the Agency in the disputes arising from them.

Tax courts, by contrast, were organized at the provincial and regional levels. Disputes arising from the audits of different Local Offices within the same province were therefore heard by the same provincial tax court at first instance and by the same regional court on appeal. Local Offices thus differed in their audit activity but not in the court system that reviewed it.

Local Office borders followed municipal boundaries, since each office served a distinct set of municipalities. However, within provinces, they were not aligned with other geographic divisions. Because several offices shared a province, they split areas that are otherwise integrated, such as local labor markets, defined by commuting patterns which often cross municipal and provincial boundaries. The same applies to other administrative units (for example, health districts below the provincial level), which are not nested in Local Offices. As a result, neighboring municipalities that belong to the same labor market, and that share the same tax court, could fall under different Local Offices. Figure 1 shows the borders of Local and Provincial Offices.

[3]The main exception concerns large taxpayers, whose audits are handled by regional offices rather than by the office of their municipality of fiscal residence.

2.1.2 The Centralization Reform

Between 2009 and 2011, the Agency implemented a centralization reform that transferred responsibility for tax controls and associated litigation from Local Offices to Provincial Offices. The reform established in each province a Provincial Directorate (*Direzione Provinciale*), which took over the competences of Local Offices in its territory. Each Provincial Directorate comprises a controls office (*Ufficio Controlli*) and one or more Territorial Offices (*Uffici Territoriali*) (Atto n. 2008/178966 del 27 November 2008). Audits and disputes that had been divided among several Local Offices within a province were thus consolidated in the controls office. The Territorial Offices, which are the former Local Offices, remained in place to provide taxpayer assistance and other front-office services. Overall, the reform replaced Local Offices as the operational units for tax controls with Provincial Directorates.

The Provincial Directorates were not all activated at once. Each was established by a separate Director's act (*provvedimento*) that set an activation date and listed the Local Offices whose tasks were transferred, so the timing of the change varies across provinces, though most adopted the new structure in 2010. Provincial Offices thereby took on operational responsibilities previously divided among Local Offices.

The reform therefore changed the administrative level at which tax enforcement and litigation were managed, consolidating these functions within provinces. Importantly, it did not alter the territorial organization of the tax courts, which continued to operate at the provincial and regional levels. This distinction allows us to examine changes in the organization of tax administration within the same judicial system.

2.2 The Italian Tax Litigation System

The Revenue Agency's organization determines which subnational office conducts tax controls and handles subsequent litigation, whereas courts follow a separate territorial structure. A formal tax dispute is typically triggered either when an audit leads the Agency to issue an assessment notice for additional tax liabilities, or when collection activity prompts a formal demand for payment. The taxpayer may either accept the claim or challenge the assessment or collection act regarding its amount, factual basis, or legality. If the taxpayer elects to litigate, the Agency becomes a party to the proceedings through the specific office that issued the act.

Before initiating judicial proceedings, taxpayers may resort to alternative dispute resolution (ADR) mechanisms designed to prevent or settle disputes. If ADR mechanisms fail, the taxpayer may formalize an appeal to the tax courts. Crucially, filing a tax dispute does not automatically suspend the underlying collection enforcement. For a contested assessment, a

taxpayer generally faces an immediate provisional payment obligation equal to one-third of the assessed tax. If the court subsequently reduces or annuls the assessment, any tax paid in excess of the amount upheld must be refunded.

Tax disputes are heard in the first instance by a provincial tax court (*Corte di Giustizia Tributaria di primo grado*) and, on appeal, by a regional tax court *(Corte di Giustizia Tributaria di secondo grado*). These courts exercise exclusive jurisdiction over disputes involving all classes of taxes, including both national and local taxes, as well as associated penalties and interest.

A crucial dimension of this system concerns the appeal rate of judgments, which provides key insights into litigation dynamics. In the period starting from 2005, this rate averaged 37% in the first instance and 21% in the second instance (Cannella et al., 2025).

Either party may appeal an adverse first-instance judgment. For the Revenue Agency, a decision that cancels or reduces a tax claim requires the litigating office to decide whether to acquiesce to the outcome or appeal to the regional court to reinstate the original claim. Before the centralization reform, this decision was made by the Local Office; afterward, it was centralized within the Provincial Office. This choice to appeal a first-instance loss constitutes the strategic office behavior studied in this paper.

In making this determination, public officials face asymmetric institutional incentives: an unappealed loss that results in the definitive cancellation of a tax claim can potentially expose the office or the individual official to liability for fiscal waste (*danno erariale*) before the Court of Auditors, thereby creating a powerful administrative incentive to pursue appeals systematically. A second-instance judgment may subsequently be challenged before the Court of Cassation strictly on questions of law. Furthermore, the financial risk of litigation for taxpayers is exacerbated by the fact that legal fees (*spese legali*) are historically offset by the tax courts, meaning they are generally not reimbursed to the taxpayer even in the event of a successful outcome.

Summary Statistics of Tax Disputes. Table 1 summarizes tax disputes handled by Local Offices in 2005–2009 and Provincial Offices in 2010–2016, using sufficiently long observation windows around the reform. The sample covers 279 Local Offices and 105 Provincial Offices. Provincial Offices handled on average 1,415 cases, of which 1,018 were first-instance cases, compared with 358 and 300, respectively, for Local Offices. This difference is consistent with Provincial Offices covering larger jurisdictions. Firms won on average 27% of first-instance cases handled by Local Offices and 31% of those handled by Provincial Offices. As discussed above, offices frequently appealed unfavorable judgments: Local Offices appealed on average 61% of first-instance judgments favorable to firms, and Provincial Offices 68%. Yet there is

substantial variation across offices. The inter-quartile range of appeal rates is 50–74% for Local Offices and 61–77% for Provincial Offices. This dispersion motivates our analysis of systematic differences in offices' propensity to appeal.

3 Data

Tax Litigation Data. We use administrative data from the Italian Ministry of Economy and Finance (MEF) covering approximately 970,000 tax disputes involving corporations between 2005 and 2022. The data identify the firms and tax authorities or public institutions involved and provide information on trial characteristics, including the court, tax type, case outcome, appeals, filing and decision dates, and amounts in dispute. The data cover proceedings at the first and second instance, but not those before the Court of Cassation.

Figure A1 illustrates the process that generates the litigation data. A dispute starts when a tax office audits a firm and may impose a sanction. The firm may then challenge the office's action before the first-instance court, which issues a decision. Disputes enter the data only when a firm brings such a challenge, so we do not observe audits or sanctions that never reach litigation. We observe all first-instance cases between firms and tax authorities. Either party that loses at first instance, whether the firm or the office, may appeal to the second-instance court, which issues a further decision. We observe all second-instance cases and which party appealed. The appeal process is illustrated in Figure A2. The highlighted box marks the office's decision whether to appeal an adverse first-instance judgment, which is the enforcement margin we study.

Firm Balance Sheet Data. We use data provided by *Cerved Group* to measure firms' performance. This dataset provides balance sheet and income statement information for the universe of Italian limited liability companies in the private non-financial sector. We link these data to the litigation records using the firm identifier, which appears in both datasets. The same identifier allows us to assign each firm to its tax office, so that we can observe the characteristics of the firms under each office's jurisdiction, including those that are never involved in litigation.

Additional Data Sources. We complement these data with the *Relazione allegata al Conto Annuale* (RELCA), an administrative survey collected by the Italian State General Accounting Department (*Ragioneria Generale dello Stato*) as part of its annual reporting on public-sector personnel. RELCA provides detailed information on staffing and the allocation of working hours across activities within individual tax offices, allowing us to measure aspects

of tax-office resources and organization that are not observable in litigation records. We construct six office-level measures: total staff (managers and non-managers) per firm in the jurisdiction, the ratio of managers to non-managerial staff, and the shares of working hours devoted to litigation, tax controls, collection, and governance. These measures combine RELCA staffing and activity records, supplemented with firm counts from Cerved. We use these measures to shed light on the organizational factors underlying differences in enforcement behavior across tax offices.

4 Heterogeneity in Tax Enforcement

4.1 Measurement

Measuring differences in tax enforcement is challenging because many dimensions of enforcement are difficult to quantify objectively, and data on them are often unavailable. By contrast, the rules of the tax system, such as tax rates, exemptions, and reporting requirements, are directly observable and have therefore received much more attention in the empirical literature (Devereux and Griffith, 1998; Suárez Serrato and Zidar, 2016; Giroud and Rauh, 2019; Fajgelbaum et al., 2019; Akcigit et al., 2021; Kennedy et al., 2026). Enforcement, by contrast, involves discretionary decisions by tax authorities, including which taxpayers to audit, which claims to pursue, and how to resolve disputes. Ideally, researchers would observe how these decisions are made. In practice, they typically observe only the outcomes of these decisions. Moreover, even outcome data are often incomplete, as they may be available only when a decision produces a particular outcome. For instance, administrative records may identify firms that were sanctioned without identifying all firms for which sanctions were considered.

Our litigation data allow us to observe the full set of decisions that each office faces on one consequential enforcement margin: whether to appeal an adverse first-instance judgment. As discussed in Section 2, when an office loses a tax dispute in first instance, it must decide whether to accept the judgment or appeal to the second-instance court. For the universe of firms covered by our analysis, we observe all such first-instance losses and whether the office appealed each judgment. Focusing on appeal decisions thus allows us to measure a well-defined dimension of enforcement that is consequential for both tax authorities and taxpayers. For the tax authority, appealing requires committing additional staff time and resources to continue pursuing the disputed tax claim. For the taxpayer, the same decision may prolong litigation, impose additional legal costs, and expose a favorable judgment to the possibility of reversal. An office's willingness to appeal is also relevant to other taxpayers assigned to it, who may face litigation in the future. Finally, to the extent that appeal behavior reflects

broader enforcement practices, its relevance may extend beyond the litigation process. We explain below how we construct this measure and then return to its interpretation.

Differences in observed appeal rates do not, by themselves, identify heterogeneity in office behavior because offices may face systematically different pools of cases. Ideally, we would compare how tax offices decide whether to appeal a given fixed case. In practice, the composition of cases that each office may appeal can differ for several reasons. First, offices' previous enforcement choices, including audits, assessments, and settlements, influence which disputes reach a first-instance judgment. Second, taxpayer characteristics differ across jurisdictions, affecting the composition of disputes, notably their financial stakes and complexity. All else equal, offices have stronger incentives to appeal higher-stakes cases. Third, the courts that different offices face may differ in their decisions and in the expected returns to appealing an adverse judgment. For instance, some courts may be more likely to rule in favor of the tax authority, affecting the composition of cases that offices lose in first instance, while appellate courts may differ in the expected probability of overturning an adverse judgment. Consequently, observed differences in appeal rates do not necessarily reflect systematic differences in enforcement behavior, as they may instead arise from differences in the cases and judicial environments that offices face. We thus measure differences in office behavior accounting for observable differences in the composition of cases and courts.

Our approach draws on both the judge-leniency and teacher value-added literatures (Chetty et al., 2014; Dobbie et al., 2018). As in judge-leniency designs, we seek to isolate a systematic decision-maker component in the propensity to choose a particular action. Unlike settings where cases are quasi-randomly assigned to judges, however, litigation cases are not randomly assigned to tax offices: taxpayer characteristics in their jurisdictions and offices' prior enforcement choices shape the cases they face. In this respect, the measurement challenge resembles teacher value-added estimation, where differences across teachers must be isolated from differences in the non-randomly assigned students they teach. Our use of the estimated propensity also differs from judge instrumental-variable designs. Rather than using it as an instrument for an individual decision, we use office aggressiveness directly as a measure of firms' enforcement exposure, exploiting quasi-exogenous changes in assigned offices induced by the centralization reform. The requirements specific to judge-leniency instruments, such as their exclusion restriction and monotonicity, therefore do not apply to our measure; identification of the effects of aggressiveness instead relies on the assumptions of our reform-based design, discussed in Section 5.1.

We measure an office's conditional appeal propensity by estimating office fixed effects while flexibly controlling for case, firm, and court characteristics. This approach relies on two advantages of our data and setting. First, we can link litigation cases to rich balance-sheet

information on the firms involved. Second, tax courts operate at a higher geographic level than tax offices, so multiple offices face the same court, which allows us to compare their appeal decisions while holding the judicial environment fixed. Specifically, we estimate the following linear probability model on the set of first-instance cases lost by tax offices:

$$A_{io} = \theta_o + X_i'\Gamma + \lambda_{c(i)\times\tau(i)} + \varepsilon_{io}, \tag{1}$$

where A_{io} indicates whether case i is appealed by office o, and θ_o denotes office fixed effects. The vector X_i includes characteristics of the litigated case and of the firm involved. Case controls account for differences in tax types, the time elapsed between the relevant tax year and the filing of the dispute, and the time between the filing of the dispute and its first-instance ruling. We additionally include a rich set of firm characteristics: total assets, revenues, profits, total debt, the debt-to-assets ratio, employment, total wage expenditures, return on assets (ROA), return on equity (ROE), firm age, industry, credit rating, and an indicator for whether the firm has exited the market. We include continuous characteristics flexibly through decile indicators, allowing for nonlinear relationships with appeal decisions. We measure these variables in the year preceding the first-instance decision. This allows us to approximate firms' economic circumstances at the time offices decide whether to appeal. Finally, $c(i)$ denotes the court ruling over case i and $\tau(i)$ the type of tax contested in case i. $\lambda_{c(i)\times\tau(i)}$ is a court-by-tax-type fixed effect, allowing appeal propensities to differ across courts and tax categories.

The estimated office effects, $\hat{\theta}_o$, capture systematic differences in appeal propensity which we interpret as a measure of aggressiveness in enforcement at the office level. Intuitively, an office with a higher estimated effect is more likely to appeal an adverse judgment when facing a case with the same observable characteristics and in the same judicial environment, reflecting a greater willingness to pursue disputed tax claims. From the taxpayer's perspective, this represents more aggressive enforcement, as a favorable first-instance judgment is more likely to be challenged, which prolongs the dispute and increases the risk of ultimately losing the case. We refer to this measure as *tax office aggressiveness* throughout the paper. At the end of this section, we provide evidence supporting this interpretation and suggesting that the measure also proxies for aggressiveness along other dimensions of enforcement behavior.

Including court fixed effects allows us to net out differences in tax court behavior, but this comes at a cost: our office effects identify only relative differences in enforcement behavior within provinces. Because tax courts operate at the provincial and regional levels, all offices operating within a province face the same courts, as discussed in Section 2. Court fixed effects therefore absorb not only differences in the judicial environment, but also any

systematic province-level component of office behavior. Consequently, we cannot separately identify differences in average aggressiveness across provinces, and the *level* of each office's aggressiveness is not identified. What we identify instead is the difference in aggressiveness among tax offices within the same province. This includes differences among Local Offices within a province, as well as between Local Offices and, after the centralization reform, the corresponding Provincial Office. We therefore normalize office effects to have mean zero within each province and interpret them exclusively in terms of within-province differences. Since our goal is to document the existence and extent of heterogeneity in enforcement within a common tax system, our measure provides a conservative assessment of this heterogeneity, since it excludes any additional variation across provinces.

4.2 Empirical Bayes Shrinkage

We apply Empirical Bayes (EB) shrinkage to reduce the influence of estimation noise in the estimates of office effects (Walters, 2024). Offices differ in the number and characteristics of the cases used to estimate their appeal propensities, so their fixed effects are estimated with different degrees of precision. For instance, offices with larger jurisdictions tend to have more cases. As a result, less precisely estimated offices may appear spuriously extreme, and part of the observed dispersion in office effects may reflect estimation noise rather than underlying heterogeneity. Empirical Bayes methods use information about the distribution of office effects to distinguish signal from noise and improve the precision of office-level estimates.

We implement parametric Gaussian EB shrinkage on the estimated office effects, after centering them with respect to each within-province mean. As Walters (2024) points out, expressing fixed effects as deviations from a common reference point mechanically introduces correlation among them. In this case, we are forced to use a common reference, the province-level mean, because of the court fixed effects included in the estimating regression. We thus account for correlated estimation errors within provinces. To do this, we first estimate the underlying dispersion of office aggressiveness by subtracting the contribution of sampling noise from the observed dispersion of estimated effects, pooling information across provinces. Each office's effect is centered relative to the unweighted mean of offices in its province. Because the office fixed effects are jointly estimated, and within-province centering introduces mechanical dependence, the estimation errors of centered office effects are correlated. We therefore use the full sampling variance-covariance matrix when estimating the underlying dispersion and calculating joint EB posterior means. The resulting estimates shrink less precisely estimated effects toward their province mean, with the degree of shrinkage reflecting the relative importance of signal and estimation noise. We denote the posterior means by

$\hat{\theta}_o^{EB}$ and use them as our preferred measure of tax office aggressiveness throughout the paper.

4.3 Results: Heterogeneity in Aggressiveness

We first document the extent of heterogeneity in enforcement across tax offices using our measure of aggressiveness. As described above, the measure captures systematic differences in the propensity to appeal adverse first-instance judgments, and accounts for differences in measurement precision across offices through Empirical Bayes shrinkage. Because the office effects are identified only relative to other offices within the same province, all the differences reported below are within-province comparisons.

We find sizeable heterogeneity in aggressiveness among offices operating within the same province. Figure 2 displays the distributions of the estimated office effects, separately for Local Offices and Provincial Offices. Both office types show considerable dispersion. This is especially visible for Local Offices: the difference between the 90th and 10th percentiles of their province-centered effects is 24.8 percentage points. On average, across provinces, the difference between the most aggressive office and the least aggressive one was 18.7 percentage points. These magnitudes are sizeable, considering an average frequency of appeals of 61%. The extent of variation across tax offices is notable given that tax rules and most enforcement guidelines are set centrally and that offices within the same province face the same courts. Substantial heterogeneity thus emerges even within a narrow geographic area and a common institutional framework.

Aggressiveness generally increased following the centralization. However, the substantial heterogeneity present before the reform meant that the changes in aggressiveness induced by the transition to Provincial Offices were also highly heterogeneous, with some areas experiencing decreases. In absolute value, the average change was 12.6 percentage points. Figure A3 plots the difference between the post-reform Provincial Office estimates and each pre-reform Local Office. The distribution is skewed toward positive changes, indicating that Provincial Offices tended to appeal more frequently, conditional on case characteristics, than the Local Offices they replaced. One possible explanation is the contemporaneous national decline in the number of first-instance tax disputes, which may have freed resources to pursue a larger share of adverse judgments through appeals (Cannella et al., 2025). Nevertheless, the distribution contains both sizeable increases and decreases.

The reform thus often implied substantially different changes for firms previously assigned to different Local Offices in the same province, as shown in Figure A4, which reports the minimum, median, and maximum changes among Local Offices within each province. The geographic distribution reinforces this result. Figure 3 maps the changes by former Local

Office jurisdiction. Positive and negative changes occur across the country, without a clear geographic gradient. Thus, because of sizeable differences at baseline, the administrative reorganization produced markedly different changes in assigned aggressiveness across firms within the same province.

4.4 Validation of the Aggressiveness Measure

We next examine whether the correlates of our measure support its interpretation as enforcement aggressiveness. Under this interpretation, offices with higher conditional appeal propensities are more willing to pursue disputed tax claims, including more marginal cases that other offices would not appeal. We examine how the measure relates to other litigation outcomes and to the organization and activities of tax offices, as reported in Figure 4. To keep these comparisons consistent with the within-province interpretation of our measure, we compute all correlations after residualizing both aggressiveness and each office characteristic on province fixed effects.

Offices with higher measured aggressiveness are also involved in tax litigation more frequently, after accounting for differences in the firms assigned to them. Figure 4 shows a positive correlation, of approximately 0.25, between aggressiveness and adjusted litigation frequency. The latter accounts for both the number and the composition of firms in an office's jurisdiction.[4] Thus, offices more willing to challenge adverse judgments are also associated with more frequent litigation among the firms they oversee, suggesting that our measure captures differences in enforcement behavior beyond the appeal decision itself.

More aggressive offices do not exhibit appreciably higher first-instance loss rates, but they are more likely to lose the appeals they pursue. Figure 4 shows only a small positive correlation between aggressiveness and the share of cases lost at first instance. Higher appeal propensity therefore does not appear to reflect offices compensating for an unusually high frequency of adverse first-instance judgments. By contrast, the correlation with the share of cases lost at second instance is substantially larger and positive. The expected probability of winning an appeal is a natural consideration in the decision to appeal: if offices appealed more frequently simply because they litigated appeals better, we would expect them to win, rather than lose, more often. Instead, the higher second-instance loss rates are consistent with more aggressive offices pursuing more marginal cases, for which the expected probability of success is lower.

[4]We measure adjusted litigation frequency using office fixed effects from a firm-year-level regression in which the dependent variable indicates whether the firm is involved in tax litigation in a given year. The regression includes the same vector of firm characteristics used in estimating office aggressiveness, allowing us to adjust for the composition of firms assigned to different offices.

Differences in aggressiveness are also associated with the organization and allocation of personnel, rather than merely with the overall size of tax offices. Figure 4 shows little relationship between aggressiveness and the number of office employees relative to the number of firms in the jurisdiction. By contrast, more aggressive offices have fewer managers per frontline worker, corresponding to a higher managerial span of control. They also devote a larger share of total working hours to litigation activities, an important consistency check for a measure based on the willingness to pursue appeals. This is accompanied by smaller shares of working hours devoted to collection and internal governance activities, while the share devoted to tax control activities is only weakly related to aggressiveness. These patterns indicate that offices with different appeal propensities also differ in how they organize their staff and allocate resources across enforcement-related tasks.

Taken together, these correlations support interpreting our measure as capturing differences in enforcement aggressiveness: offices with higher conditional appeal propensities appear to pursue more marginal appeals, rather than simply responding to more first-instance losses or benefiting from superior litigation performance. The associations with litigation frequency and the allocation of office personnel further suggest that the measure of aggressiveness captures behavior that extends beyond appeal decisions.

5 The Impact of Tax Office Aggressiveness on Firms

Simply regressing firm outcomes on the measured aggressiveness of the assigned tax office would not generally identify its causal effect. Since firms are assigned to offices according to their location, differences in aggressiveness may be correlated with local economic conditions or firm characteristics that independently affect outcomes. Moreover, the composition and performance of firms in an office's jurisdiction may themselves influence the office's propensity to appeal adverse decisions, generating reverse causality. We address these concerns with our empirical strategy.

5.1 Identification Strategy

The centralization reform described in Section 2 provides variation in the aggressiveness of the tax office to which firms are assigned. Before the reform, firms in the same province faced different Local Offices, each responsible for a distinct set of municipalities and characterized by its own level of aggressiveness. The jurisdictions of Local Offices consisted of geographic clusters of municipalities and clusters were idiosyncratic to the tax authority. Their boundaries partitioned provinces and did not generally coincide with those of other local institutions,

including civil courts (Giacomelli and Menon, 2017) or the areas served by schools and hospitals. After centralization, all firms within a province were assigned to the same Provincial Office and faced a common level of aggressiveness. Hence, within each province, the sign and size of the change in aggressiveness caused by the reform were determined by the aggressiveness of each firm's pre-reform Local Office. The reform thus allows us to combine cross-sectional variation in baseline aggressiveness across geographically granular Local Office jurisdictions with variation in aggressiveness over time within those same jurisdictions.

To illustrate the identification strategy, consider a single province with two Local Offices at baseline, l' and l'', and two periods, before ($t = 1$) and after ($t = 2$) the reform. Let $a_{l'}$ and $a_{l''}$ be their pre-reform aggressiveness, and a_p the aggressiveness of the common Provincial Office after the reform. For firm i, originally assigned to office $l(i)$, the reform-induced change is

$$D_i \equiv a_p - a_{l(i)} \in \{d', d''\}, \qquad d' = a_p - a_{l'}, \quad d'' = a_p - a_{l''}. \tag{2}$$

We assume $d' \neq d''$ and neither equals zero. This mirrors our empirical setting, where aggressiveness differs across Local Offices at baseline and between Local and Provincial Offices. As explained in Section 4.1, only within-province differences in aggressiveness are identified. For this reason, we focus on identification within a single province. The estimation strategy below pools these comparisons across provinces and there we discuss how we interpret the pooled estimates.

Potential Outcomes Framework. Consider firm i with pre-reform aggressiveness equal to $a_{l(i)}$. We denote as Y_{it} its *observed* outcome in period t, and as $Y_{it}(d)$ its *potential* outcome in period t under a reform-induced change in aggressiveness of d. We denote as $\mathcal{D}$ the set of possible changes in aggressiveness.[5] In particular, $Y_{it}(0)$ is the potential outcome that would be observed in the counterfactual in which aggressiveness did not change around the reform. That corresponds to $a_p = a_{l(i)}$.

This formulation is implicitly assuming the Stable Unit Treatment Value Assumption (SUTVA), which we maintain. This assumption requires that firm i's potential outcome $Y_{it}(d)$ is only a function of its own treatment assignment. Additionally, we assume that $Y_{i1}(d) = Y_{i1}(0)$, for all $d \in \mathcal{D}$. That is, future changes in aggressiveness do not affect pre-reform potential outcomes. This is the usual assumption of 'no anticipation' in standard difference-in-differences (DID) settings, applied to our setting with a continuously-valued treatment.

For firms that experience a change in aggressiveness equal to d, we define the average

[5]Since we measure aggressiveness empirically as the office-level propensity to appeal adverse tax litigation outcomes, which ranges from 0 to 1, in principle the set is $\mathcal{D} = [-1, 1]$.

treatment effect of this change as

$$ATT(d \mid d) \equiv E[Y_{i2}(d) - Y_{i2}(0) \mid D_i = d]. \tag{3}$$

We condition on the observed treatment d because, in principle, average treatment effects could be heterogeneous across observed treatment realizations. We use the difference in potential outcomes *after* the reform, $t = 2$. With more post-reform periods, we could define a different $ATT_t(d \mid d)$ for each period t. We will return to this when discussing the dynamics of the effects, but for now we focus on the case with one post-reform period for simplicity.

More generally, since aggressiveness and its change d are continuous, we are interested also in the entire average treatment-effect function. For a hypothetical change $h \in \mathcal{D}$,

$$ATT(h \mid d) \equiv E[Y_{i2}(h) - Y_{i2}(0) \mid D_i = d], \qquad \forall\ d \in \mathcal{D}. \tag{4}$$

For instance, we may be interested in the marginal effect of a change in aggressiveness around the observed change d:

$$\tau(d) \equiv \left.\frac{\partial ATT(h \mid d)}{\partial h}\right|_{h=d}. \tag{5}$$

We now discuss the assumptions that allow us to recover these causal parameters from the observed variation in outcomes and change in tax office aggressiveness.

Parallel trends. The first requirement is that, within each province and absent changes in aggressiveness, firms initially assigned to different Local Offices would have experienced on average the same trend in outcomes.

Assumption 1 (Parallel trends across initial aggressiveness levels)**.** *For any $d', d'' \in \mathcal{D}$,*

$$E[Y_{i2}(0) - Y_{i1}(0) \mid D_i = d'] = E[Y_{i2}(0) - Y_{i1}(0) \mid D_i = d'']. \tag{6}$$

Let $\Delta Y_i = Y_{i2} - Y_{i1}$ and $B(d) \equiv E[Y_{i2}(0) - Y_{i1}(0) \mid D_i = d]$. ΔY_i is the change in observed outcomes across periods. $B(d)$ is the average change in outcomes that we would have observed among firms experiencing a change in aggressiveness equal to d, had the reform not happened and had they remained at the level of aggressiveness of their local office $a_{l(i)}$.

No anticipation implies $E[\Delta Y_i \mid D_i = d] = ATT(d \mid d) + B(d)$. Under Assumption 1, the

within-province difference-in-differences (DID) comparison satisfies[6]

$$\begin{aligned} DID(d'', d') &\equiv E[\Delta Y_i \mid D_i = d''] - E[\Delta Y_i \mid D_i = d'] \\ &= ATT(d'' \mid d'') - ATT(d' \mid d'). \end{aligned} \tag{7}$$

Initial aggressiveness may affect the *levels* of firm outcomes; the assumption requires only that such differences would not generate differential changes absent the reform-induced variation in aggressiveness. In particular, dynamic effects of initial aggressiveness on counterfactual outcome trends would violate this restriction. Hence, we need to assume that any impact of the initial level of aggressiveness had stabilized to a constant gap between firms in different Local Offices' jurisdictions. We believe this is plausible given that Local Offices had been active long before the reform, since the creation of the Italian Revenue Agency. We discuss how pre-trend differences can inform us on the plausibility of this assumption when we discuss our empirical specification.

Assumption 1 alone identifies the difference between the effects on two groups receiving different changes, not a common causal response to changes in aggressiveness. This means that, without additional assumptions, the quantity identified by $DID(d'', d')$ remains hard to interpret causally. Still, $DID(d'', d') \neq 0$ implies that at least one of the two causal effects, $ATT(d'' \mid d'')$ and $ATT(d' \mid d')$, differs from 0. So it already provides a first answer to the question of whether differences in aggressiveness affect firm outcomes.

Treatment-effect heterogeneity. In more standard continuous-treatment settings, one would compare units that begin at the same initial level of aggressiveness and only some of them experience a change in it.[7] Here every firm changes the level of aggressiveness it faces, and different changes necessarily correspond to different initial aggressiveness levels. We therefore need to discipline how causal effects change across observed treatment assignments. We do so by imposing the following restriction on selection into gains from treatment.

Assumption 2 (Common average effects)**.** *For any change $h \in \mathcal{D}$ and any two observed changes d', d'',*

$$ATT(h \mid d') = ATT(h \mid d''). \tag{8}$$

[6]We refer to this comparison as difference-in-differences because it involves the difference of before-after differences between two groups. However, it should not be understood as a standard DID where one group experiences no treatment and acts as a natural comparison group.

[7]A recent literature on difference-in-differences with continuous treatments discusses the (weaker) assumptions that one can make in these more standard settings (Callaway et al., 2024; Chaisemartin, d'Haultfoeuille, et al., 2025). In particular, Chaisemartin, D'Haultfœuille, et al. (2024) extend the framework to a setting in which all units experience a change in treatment but for some the change is *negligible*. This is not the case in our setting.

We assume that the average effect of a given change in aggressiveness is therefore the same for firms with different baseline levels of aggressiveness. Individual effects may differ and the average effect may vary with the size of the change and across provinces. In other words, we allow differences in baseline aggressiveness to influence treatment effects, but only to the extent that they determine the change in aggressiveness.

For instance, say that Local Office A has a lower level of aggressiveness than Local Office B. Assumption 2 allows the actual average treatment effect to be larger for firms assigned to A, who thus experience a relatively larger increase. But it assumes that firms assigned to B would have experienced the same average effect, had they had the same change in aggressiveness that firms assigned to A had.

Assumption 2 rules out selection on average gains associated with the initial office assignment. This assumption may be violated if the composition of firms in Local Offices' jurisdiction was affected by the offices' aggressiveness and if different firms would experience different effects of the same change in the aggressiveness they face. We can partially relax this assumption by comparing firms with similar observable characteristics. For instance, we can compare firms within the same sector. More problematic is selection along unobserved characteristics. This can happen along three margins: entry, exit, and relocation of firms across different local offices. We can exclude the third margin in practice. In the time period we consider, a negligible fraction of firms ever relocates to an area that falls under a different Local Office jurisdiction and we exclude these firms from the sample. While we cannot rule this out for entry and exit, we note that, if this were the case, and these effects were sufficiently large to severely bias our DID comparisons, this would at minimum prove that differences in aggressiveness have important consequences for two main margins of economic activity.

Under Assumptions 1 and 2, the average effect of a given change in aggressiveness is common across groups with different observed changes. We can therefore define the $ATT(h) \equiv ATT(h \mid d)$, independently of d. Equation 7 then becomes

$$DID(d'', d') = ATT(d'') - ATT(d').$$

Thus, the DID comparison identifies differences in the average treatment-effect function across observed changes in aggressiveness. In particular, if we divide $DID(d'', d')$ by the difference $d'' - d'$, the ratio identifies the average slope of the treatment-effect function between d' and d''. If the ATT function is differentiable, for sufficiently small differences between d' and d'', this ratio approximates the marginal causal effect $\tau(d')$ defined in Equation 5.

However, these assumptions are not sufficient to identify the average treatment effect

$ATT(d)$ itself, or the entire treatment-effect function. Since no firm experiences $d = 0$, while we can identify differences in $ATT(d)$ across observed changes, we cannot recover its level relative to the no-change counterfactual. To identify these additional causal parameters, we need stronger restrictions on how average treatment effects vary with the change in aggressiveness.

Functional form. To anchor the dose-response function at the no-change counterfactual, we adopt as a natural baseline a linear restriction.

Assumption 3 (Linear average treatment effects)**.** *The common average treatment-effect function satisfies*

$$ATT(d) = \beta d \tag{9}$$

for each change $d \in \mathcal{D}$.

Combining Assumptions 1, 2, and 3 yields

$$\tau(d) = \beta = \frac{DID(d'', d')}{d'' - d'}, \qquad d'' \neq d'. \tag{10}$$

Thus, under this additional restriction, the DID comparison identifies the entire average treatment-effect function, $ATT(d) = \beta d$.[8]

5.2 Empirical Specification

We estimate the relationship using the firm-year panel, pooling provinces while retaining within-province variation:

$$Y_{it} = \beta_P \Delta\hat{\theta}^{EB}_{l(i)} \times P_t + \beta_T \Delta\hat{\theta}^{EB}_{l(i)} \times T_t + \mathbf{X}'_{m(i)t}\boldsymbol{\gamma} + \alpha_i + \lambda_{p(i)t} + \varepsilon_{it}. \tag{11}$$

Here Y_{it} is an outcome of firm i in year t; $l(i)$ denotes its pre-reform Local Office and $p(i)$ its province. The reform-induced change in aggressiveness of the assigned tax office is $\Delta\hat{\theta}^{EB}_{l(i)} \equiv \hat{\theta}^{EB}_{p(i)} - \hat{\theta}^{EB}_{l(i)}$. It is the difference between measured aggressiveness of the post-reform Provincial Office and of the pre-reform Local Office, with Empirical Bayes (EB) shrinkage applied to each measure. Since all firms assigned to a Local Office get assigned to the same Provincial Office after the reform, the change varies at the level of the Local Office jurisdiction. The indicators T_t and P_t equal one respectively in the transition period (2009–2011) and

[8]With K distinct observed changes within a province, a polynomial average response $ATT(h) = \sum_{j=1}^{q} \beta_j h^j$ is identified, together with an unknown common counterfactual trend, when $K \geq q + 1$. The two-office example therefore motivates a linear specification, but richer variation permits more flexible functional forms.

post-reform period (2012–2019). We omit the pre-reform period for which our data is available, 2005–2008, as reference. The transition period accommodates the staggered implementation of centralization across provinces, which began in 2009 and was completed during 2011. Our coefficient of interest is β_P. β_T allows for a distinct relationship during the transition.

The vector $\mathbf{X}_{m(i)t}$ includes controls at the level of firm i's municipality $m(i)$ for population, local taxation, and state capacity. Specifically, we control for population and its square; deductions for the principal residence and ordinary and principal-residence tax rates under ICI, IMU, and TASI; and the three components of the Municipal Administrative Quality Index (MAQI), measuring the quality of the municipal bureaucracy, the quality of local politicians, and municipal fiscal and economic performance (Cerqua et al., 2024). These controls are particularly important because treatment is assigned at the level of Local Office jurisdictions, each consisting of a group of municipalities. Firm fixed effects α_i absorb time-invariant differences across firms and their initial Local Office jurisdictions. Province-by-year fixed effects $\lambda_{p(i)t}$ absorb common provincial shocks and province-wide consequences of the reform. We cluster standard errors at the level of our variation, jurisdictions of Local Offices.

The coefficient β_P is identified by differences in outcome changes across firms initially assigned to different Local Offices in the same province. Province-by-year fixed effects also ensure that estimation uses only within-province variation, as required by the construction of our aggressiveness measure.[9] We scale aggressiveness measures by 10 percentage points for ease of interpretation. This is an easily interpretable benchmark and close to the magnitude of the observed changes in aggressiveness induced by the reform. In absolute value, the mean change was 12.6 percentage points and the median 9.7. Hence, under Assumptions 1, 2, and 3, β_P measures the effect of a 10-percentage-point increase in aggressiveness on outcome Y in the post-reform period.

As discussed when constructing our measure, the propensity to appeal is intended to capture broader differences in tax-office aggressiveness. To the extent that other enforcement practices change together with appeal propensity, these changes are part of the enforcement variation we seek to capture. Other consequences of centralization threaten this interpretation only if they affect firm outcomes differentially across initial Local Office jurisdictions in ways correlated with $\Delta\hat{\theta}^{EB}_{l(i)}$. Importantly, common province-wide changes are absorbed by $\lambda_{p(i)t}$. Finally, we estimate an event-study version of equation (11), interacting $\Delta\hat{\theta}^{EB}_{l(i)}$ with years relative to the reform. We use pre-reform coefficients to assess whether firms facing different subsequent changes were already on differential trends. Post-reform coefficients instead trace

[9] In principle, we could estimate a separate coefficient $\beta_{P,p}$ for each province. We pool provinces to have more power. If the marginal effects $\beta_{P,p}$ differ across provinces, the pooled coefficient summarizes these province-specific effects, with weights depending on sample size and identifying variation.

how the relative effects evolve over time.

5.3 Main Sample

Our main sample includes firms located in areas for which the reform generates within-province variation in measured tax-office aggressiveness. Specifically, we retain firms assigned to Local Offices for which both pre-reform Local Office and post-reform Provincial Office aggressiveness can be estimated, and where more than one Local Office operated within the province before the reform. The resulting sample comprises firms assigned to 255 Local Offices across 83 provinces. We exclude 22 provinces with only one Local Office at baseline, for which there is no within-province variation in treatment, and Aosta, where Provincial Office aggressiveness is not measured.[10]

Table 2 reports summary statistics for the main sample over 2005–2019. The table focuses on the firm outcomes analyzed in Section 5.5 to estimate the effects of tax-office aggressiveness. On average, 6% of firms exit in a given year, revenues are approximately 1.6 million euros, firms employ 7 workers, and return on assets (ROA) is 3.5%. We also report tax litigation frequency, which is relatively low across all firms. Over 2005–2016, there are on average 17 first-instance cases per thousand firms per year; 5 of these end in judgments favorable to the firm, and the tax office appeals 3 of them.

5.4 Expected Effects and Channels

Our goal is to establish whether the differences in tax-office aggressiveness documented in Section 4 have an impact on firm outcomes. The tax-enforcement literature has traditionally focused on firm compliance and tax revenues (Slemrod, 2019), but its findings also suggest channels through which enforcement may affect firms' real activity. We use these channels to guide our analysis. We group them into three: the administrative and financial burden of enforcement, firms' incentives to comply, and spillovers across firms.

First, more aggressive enforcement can increase the *administrative and financial burden* of dealing with the tax authority. An office more willing to appeal unfavorable judgments can prolong disputes, requiring additional legal and accounting services and diverting managerial time, even when the firm ultimately prevails. Indeed, Cannella et al. (2025) find that firms undergoing tax disputes experience a decline in financial performance compared to similar firms, even when they win the dispute. More broadly, stricter enforcement may increase the likelihood of audits, contested assessments, and sanctions. Tax compliance costs are

[10]In Aosta, the reform led to a different organizational structure, in which the Regional Office also handled litigation activities. We therefore exclude it from the sample.

substantial, even for large firms (Blumenthal and Slemrod, 1996). Uncertainty about future tax liabilities can also induce precautionary cash holdings and discourage investment (Hanlon et al., 2017; Jacob et al., 2022; Cowx, 2025). Consistent with these costs having real consequences, Belnap et al. (2024) find that random tax audits reduce survival, with some evidence of an additional administrative-cost channel. These effects may be particularly important for firms with limited liquidity or greater exposure to disputes. Hence, *ex ante*, which types of firms will be more affected by a change in aggressiveness of enforcement is ambiguous. Smaller firms may be less able to sustain increased costs. At the same time, because of size-dependent enforcement, larger firms may be more exposed to them.

Second, aggressiveness may change firms' *incentives to comply*, even before an audit or dispute occurs. In the classic framework of Allingham and Sandmo (1972), greater expected detection and penalties reduce the return to evasion. Evidence from random audits shows that taxpayers adjust their reported income after receiving stronger enforcement signals (Slemrod et al., 2001; Kleven et al., 2011). Monitoring and information trails, which increase the probability of detection, also affect firms' reporting and avoidance decisions (Pomeranz, 2015; Almunia and Lopez-Rodriguez, 2018; Hoopes et al., 2012). For non-compliant firms, increasing tax compliance can reduce after-tax cash flow and the resources available for hiring or investment, producing negative real effects. Yet observed changes in revenues and profits do not necessarily correspond to changes in real activity. For instance, Carrillo et al. (2017) find that firms responding to detected revenue discrepancies increased reported revenues but also offset much of the adjustment with higher reported costs. Hence, while the literature predicts negative *real* effects of stronger enforcement on non-compliant firms, reporting behavior may change observed outcomes in the opposite direction. Our balance-sheet measures of firm performance will capture the net effect of these two forces.

Third, enforcement may generate *spillovers* between more and less directly affected firms.[11] Firms that evade taxes effectively operate with a cost advantage over otherwise similar competitors. Stricter enforcement that reduces this advantage could shift sales and production toward more compliant or less-exposed firms. Di Marzio et al. (2026) show that noncompliance reduces competitors' revenues and productivity, illustrating the potential importance of this margin. Enforcement can also propagate through business relationships: Pomeranz (2015) finds compliance spillovers along VAT supply chains, while Boning et al. (2020) document effects transmitted through shared tax preparers. However, spillovers need

[11]Strictly speaking, spillovers across firms would violate the SUTVA under the firm-level potential-outcomes notation used for simplicity in our identification framework. However, the reform-induced change in aggressiveness varies at the level of pre-reform Local Office jurisdictions, rather than at the level of individual firms. Defining potential outcomes with respect to jurisdiction-level treatment allows for spillovers among firms within the same jurisdiction, which are part of the effects we aim to capture.

not operate only through differences in tax compliance. If some firms are more vulnerable to the administrative and financial costs of enforcement, they may contract or exit, allowing less affected competitors to expand even in the absence of differences in compliance behavior. These indirect responses depend on firms' relative exposure and market relationships and therefore need not benefit every less-exposed firm.

5.5 Main Results

We examine four outcomes capturing different dimensions of firm performance. On the extensive margin, we measure firm survival using an indicator that equals one if a firm has exited by year t, and remains equal to one thereafter, thus capturing cumulative exits. On the intensive margin, we consider two measures of business activity, revenues and employment, and a standard measure of profitability, return on assets (ROA). For these outcomes, we restrict the sample to active firms. Hence, when aggressiveness affects firm survival, changes in these outcomes may also reflect changes in the composition of surviving firms.

For the average firm in our data, increased tax-office aggressiveness is associated with lower revenues and profitability. Table 3 reports the effects of a change in measured tax-office aggressiveness in the transition and post-reform periods, using all firms. In the post-reform period, a 10-percentage-point increase in aggressiveness is associated with a 1.2% decline in revenues and a 0.1-percentage-point decline in ROA, about 3.8% in relative terms, both statistically significant at the 10% level. The estimated effects on cumulative exits and employment are close to zero and statistically insignificant. During the transition period, the coefficients are generally small and statistically insignificant, with the exception of a modest positive effect on employment. Overall, the estimates suggest negative average effects on revenues and profitability, although they are relatively small and imprecisely estimated. As we show below, these average effects conceal substantial heterogeneity across firms and industries.

Next, we examine the dynamics of these effects, estimating the event-study analogue of specification (11) and omitting 2008 as the reference year. Figure 5 reports the estimates for each outcome. The pre-reform coefficients are generally close to zero and do not exhibit systematic trends, which supports Assumption 1 of parallel trends for firms initially assigned to different Local Offices. Following the reform, the effects on revenues and ROA become negative and persist throughout the post-reform period. In contrast, employment exhibits a modest positive response in the early years following the reform, which fades in the later period. The effect on cumulative exits increases gradually over time, albeit imprecisely estimated. This pattern is consistent with worsening firm performance, as shown by revenues

and profitability, affecting firm survival with a lag. Overall, the event studies suggest that the negative effects of higher aggressiveness on revenues and profitability emerge quickly, while the effects on survival take longer to materialize.

While the average firm experiences relatively modest declines in performance when aggressiveness increases, the effects are much larger for firms less able to withstand negative shocks. To show this, we estimate the effects of the reform-induced change in aggressiveness separately for firms with limited liquidity. Specifically, we classify as "highly liquidity-constrained" those firms whose average current liquidity ratio (i.e., current assets divided by current liabilities) is below the 25th percentile of the distribution. We take the average value over pre-reform years, 2005–2008, to avoid capturing the impact of the change in aggressiveness on this measure. We then estimate a saturated version of specification (11), allowing the coefficients on changes in aggressiveness to differ between highly and less liquidity-constrained firms in both the transition and post-reform periods, and allowing for group-specific period effects.

Table A1 shows that the negative effects of aggressiveness are concentrated among highly liquidity-constrained firms. For these firms, a 10-percentage-point increase in aggressiveness raises cumulative exit by 0.9 percentage points and reduces revenues and employment by 4.0% and 0.8%, respectively. All three estimates are large and statistically significant. ROA also declines by a sizeable 0.2 percentage points, but the estimate is imprecise. Less liquidity-constrained firms, by contrast, are largely unaffected. The corresponding coefficients are much smaller and, although generally precisely estimated, none is significant. This pattern is consistent with the costs of more aggressive enforcement being particularly consequential for firms with limited liquidity.

We also document that, for smaller firms, an increase in aggressiveness leads to a worsening of performance, while for larger firms effects are mixed. We proxy size using average revenues over the pre-reform period. Firms whose average revenues are in the top quartile of the distribution are classified as large, and we estimate effects separately for large firms and all other firms. *A priori*, as discussed in Section 5.4, heterogeneity along this dimension is ambiguous. Large firms may be less affected because they generally engage less in evasion and are better able to cope with administrative burdens and temporary shocks, but they may be more affected if they are disproportionately targeted by the tax authority because of size-dependent enforcement (Almunia and Lopez-Rodriguez, 2018). We find that smaller firms experience substantially more adverse effects (Table A2). Among smaller firms, a 10-percentage-point increase in aggressiveness raises cumulative exit by 0.3 percentage points, reduces revenues by 3.2% and employment by 0.6%, and lowers ROA by 0.13 percentage points. All four estimates are statistically significant. For larger firms, the effect on employment is, if

anything, positive and statistically significant, a 1.3% increase, while the other effects are not statistically distinguishable from zero. Together with the liquidity results, this pattern suggests that smaller and financially vulnerable firms are disproportionately affected by more aggressive enforcement.

Finally, we examine heterogeneity across economic sectors. Figure 6 reports post-reform coefficients estimated separately by sector.[12] The most adverse responses tend to be concentrated in services. In accommodation and food services, a 10-percentage-point increase in aggressiveness raises the cumulative exit probability by 1.3 percentage points and reduces revenues by about 2.6%. Revenues also decline by 4.6% in other services, while profitability falls in professional services and rental and support services. In contrast, the estimated effects in manufacturing are generally smaller and statistically insignificant. Notably, several of the most affected industries are also characterized by relatively high rates of irregular employment and tax noncompliance (Riccardi et al., 2018). This is consistent with the possibility that differences in opportunities for evasion, and hence in responses to deterrence, contribute to the heterogeneity across sectors.

These differences across sectors partly reflect differences in composition by liquidity constraints and firm size, but not entirely. Appendix Figures A5 and A6 report sector-specific effects separately along these two dimensions. Once we separate firms by liquidity constraints and size, heterogeneity across industries generally becomes smaller. Within sectors, highly liquidity-constrained firms tend to experience more adverse effects, while smaller firms tend to reduce revenues and employment and larger firms often show the opposite pattern. However, in the most affected sectors, negative effects are also present among less-liquidity-constrained and larger firms. For example, the increase in exit among firms in accommodation and food services remains large and significant even among less-liquidity constrained and larger firms.

Magnitude of the effects. The magnitude of our estimates suggests that tax-office aggressiveness affects firms through broader enforcement practices and behavioral responses, rather than only through the direct consequences of additional appeals. To assess how much of the effect can be explained by additional appeals alone, we construct a back-of-the-envelope upper bound. Consider the frequency of appeal decisions from the point of view of a firm. Focusing on liquidity-constrained firms, our data show that there are on average 0.004 favorable first-instance judgments per year for these firms.[13] A 10-percentage-point increase in the probability that the tax office appeals these judgments generates relatively few

[12]We group firms by their NACE level 1 sector code. This is comparable to one-digit industry codes in the SIC classification.

[13]In the pre-reform period, liquidity-constrained firms had on average 0.017 first-instance tax cases per firm-year, approximately 26% of which resulted in a judgment favorable to the firm.

additional appeals. Allowing their effects to accumulate over time, even under the extreme assumption that every additional appeal causes the firm to exit, the implied increase in cumulative exit averages at most 0.29 percentage points over the post-reform period. That is, about one-third of our estimated 0.9-percentage-point effect. In practice, the effect of an additional appeal on the probability of exit is likely to be substantially smaller.[14] This finding is consistent with the evidence that appeal propensity proxies for broader enforcement practices.

More aggressive tax offices may also audit firms more frequently. We assess the potential contribution of this channel using estimates from the literature on the effects of tax audits. Belnap et al. (2024) find that an audit increases the probability of closure for small businesses by 1.2–6.0 percentage points, depending on the measure, and reduces subsequent revenues among surviving firms by approximately 22%. Suppose that a 10-percentage-point increase in measured appeal propensity also reflects a 10-percentage-point increase in the probability of being audited. Applying these estimates suggests that additional audits could explain approximately 13–67% of our estimated effect on exit and 55% of the decline in revenues among liquidity-constrained firms.[15] These benchmarks suggest that broader enforcement practices can generate effects of the magnitude we estimate, even before accounting for firms' additional behavioral responses.

Economic channels. The heterogeneity across firms and industries provides further insights into the channels through which tax-office aggressiveness affects economic activity. First, the larger effects on liquidity-constrained firms point to the importance of administrative and financial costs, including the direct expenses of dealing with tax authorities and broader compliance burdens (Kalmenovitz, 2023; Trebbi et al., 2026). Second, more aggressive enforcement increases the expected costs of noncompliance, with disproportionately higher effects where opportunities for tax evasion are greater. Non-compliant firms may respond by changing their reporting behavior, incurring additional compliance costs, or adjusting real economic activity. For marginal firms whose survival depends on tax evasion, stronger enforcement may also induce exit. The concentration of adverse effects in industries with higher measures of tax noncompliance, especially accommodation and food services, is consistent with this deterrence channel. Finally, our findings also suggest a role for competitive spillovers. Within industries, where smaller firms tend to experience declines in revenues and employment, larger firms often exhibit positive responses along these margins. This pattern is consistent

[14] Cannella et al. (2025) show that firms undergoing tax litigation are 2 percentage points more likely to exit than comparable firms, and 3.5 percentage points more likely if their cases are appealed.

[15] We multiply the estimated effects of an audit by the assumed 10-p.p. increase in audit probability and divide by our estimates.

with economic activity being reallocated toward larger firms as smaller competitors contract or exit in response to more aggressive enforcement. Moreover, because tax evasion and irregular employment are disproportionately concentrated among smaller firms, the positive responses of larger firms are less likely to reflect changes in reporting behavior and more likely to represent real expansions in economic activity (Riccardi et al., 2018). Overall, the evidence points to a combination of enforcement-related costs, deterrence, and competitive reallocation in shaping firms' responses to tax-office aggressiveness.

6 Conclusion

This paper shows that the implementation of a common tax system can differ substantially across local authorities and that these differences have meaningful consequences for firms and economic activity. We develop a novel measure of tax-office aggressiveness based on offices' propensity to appeal adverse first-instance judgments, adjusting for case, firm, and court characteristics. We document substantial heterogeneity even among offices operating within the same province, under common tax rules and facing the same tax courts. The evidence suggests that the measure captures a broader dimension of aggressiveness: offices with systematically higher appeal propensities pursue more marginal appeals, are involved in more litigation cases than expected given the firms in their jurisdictions, and devote a larger share of staff working hours to litigation.

To examine the consequences for firms, we exploit a reform that centralized tax enforcement from Local to Provincial Offices, generating heterogeneous changes in the aggressiveness of the offices assigned to firms. Using within-province comparisons in a continuous difference-in-differences design, we find that greater aggressiveness reduces firm revenues and profitability on average, although aggregate effects are relatively modest. The negative effects are substantially larger among smaller and liquidity-constrained firms. For liquidity-constrained firms, a 10-percentage-point increase in aggressiveness reduces revenues by 4% and employment by 0.8%, and raises the probability of exit by 0.9 percentage points. By contrast, larger firms seem largely unaffected and, if anything, increase their employment. We also find stronger adverse effects in service sectors, which offer more opportunities for tax evasion. These patterns are consistent with a combination of higher administrative and financial burdens, stronger incentives for tax compliance, and reallocation of economic activity across more *versus* less directly affected firms.

Overall, our findings show that differences in the implementation of tax rules, even within a common national tax system, are substantial. Moreover, these differences can generate meaningful differences in the economic environment firms face. The implication is that

understanding the economic consequences of taxation requires considering not only statutory rules, but also how tax authorities exercise discretion in enforcing them.

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Main Figures & Tables

Figure 1: Borders of Local, in **red**, and Provincial Offices, in **black**

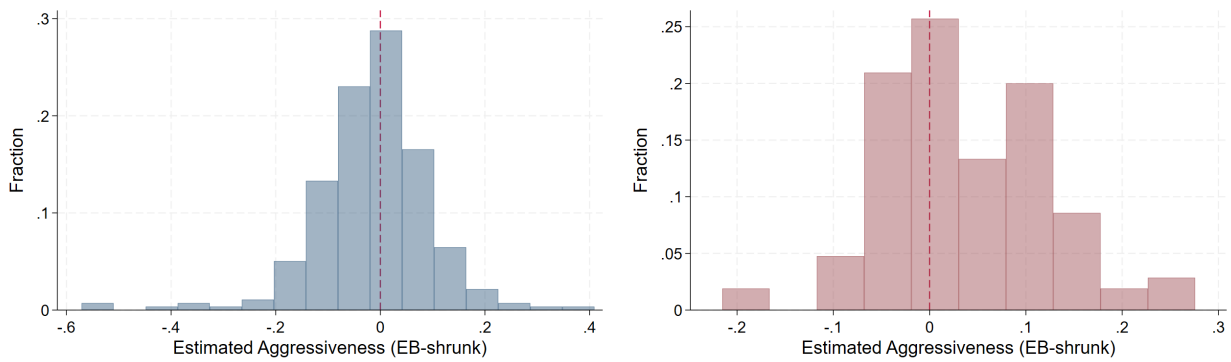


Notes: The map shows the geographic boundaries of local and provincial tax office jurisdictions. Red lines indicate local office boundaries, and **black** lines indicate provincial office boundaries.

Figure 2: Office-Level Centered EB Aggressiveness

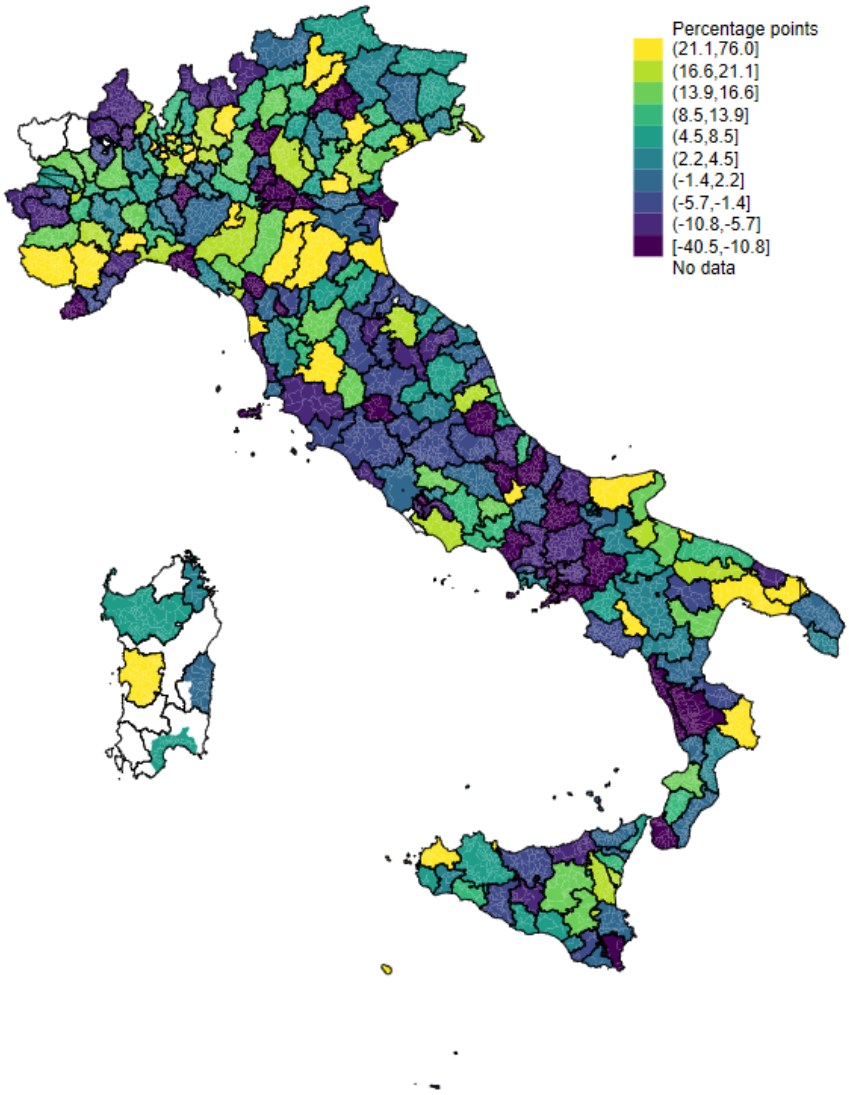
(a) Local Offices

(b) Provincial Offices



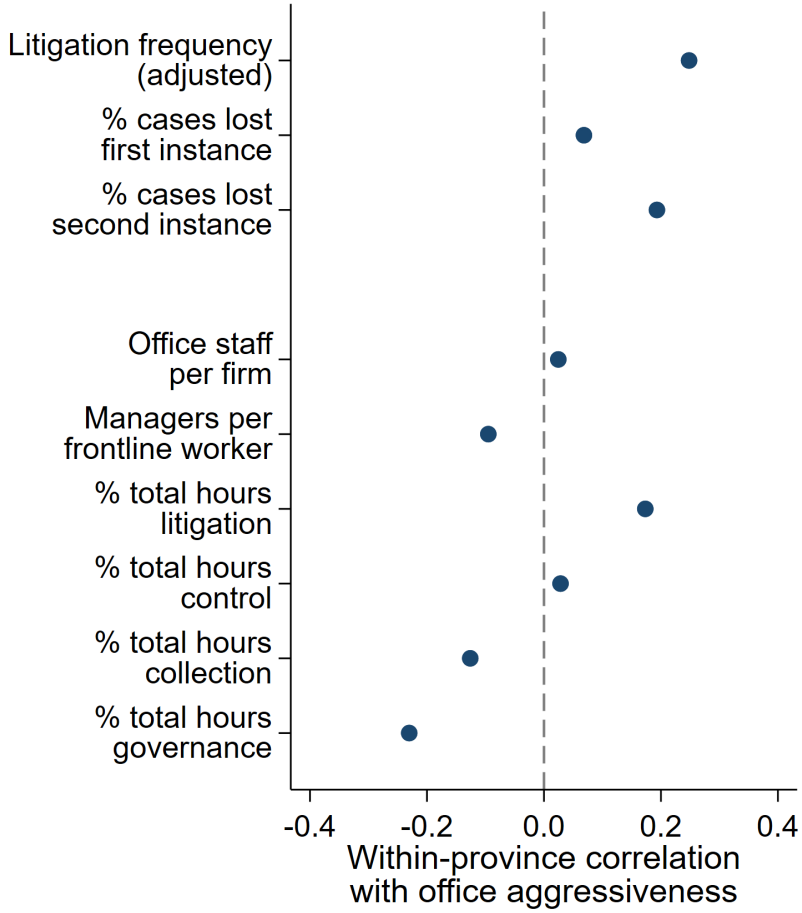
Notes: The figure shows the distribution of estimated appeal aggressiveness across local tax offices (panel a) and provincial offices (panel b). Estimates are empirical Bayes–shrunk office effects, centered within province. Each office with a non-missing estimate contributes one observation. The vertical line marks zero.

Figure 3: Change in Tax Office Aggressiveness by Local Office Jurisdiction



Notes: The map shows the difference between estimated provincial and local office appeal aggressiveness assigned to each municipality, expressed in percentage points. Estimates are empirical Bayes–shrunk office effects, and the difference is defined as provincial minus local aggressiveness. Positive values indicate that the provincial office is more aggressive than the corresponding local office. Black lines mark local office boundaries.

Figure 4: Correlates of Tax Office Aggressiveness Measure



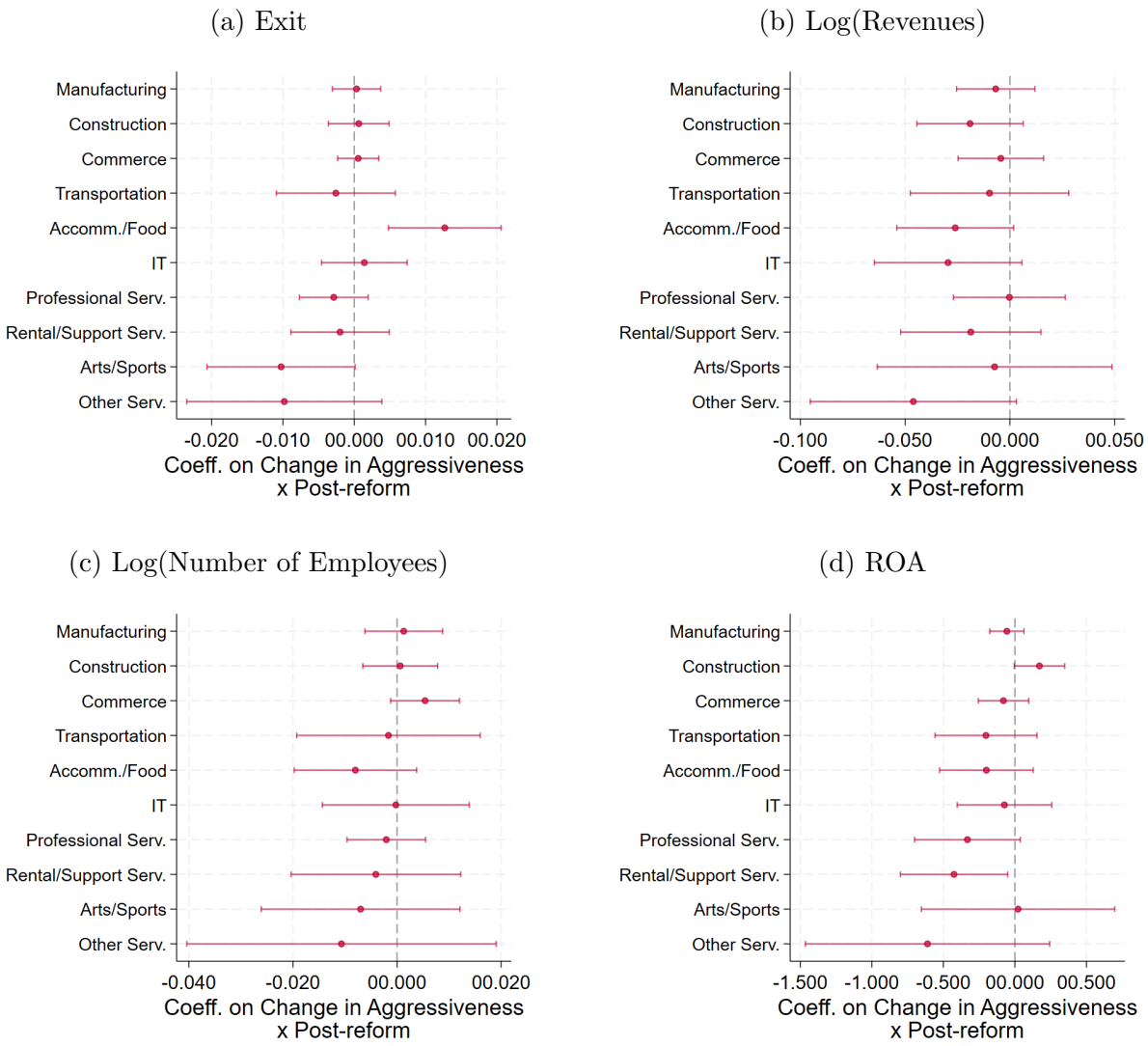
Notes: The figure reports within-province correlations between estimated office appeal aggressiveness and the office characteristics listed on the vertical axis, using local and provincial offices. Both the characteristic and the empirical Bayes–shrunk aggressiveness are demeaned within province using their shared non-missing sample, in order to remove province-level differences. The dots show correlations between the resulting deviations from province means.

Figure 5: Event-Study Estimates



Notes: The figures report coefficients on the interaction between year indicators and the reform-induced change in estimated appeal aggressiveness, defined as provincial minus local office aggressiveness and measured in units of 10 percentage points. This is the event-study analogue of specification (11). Outcomes are indicated by the panel caption. The omitted year is 2008. Regressions include firm and province-by-year fixed effects and municipality controls. Standard errors are clustered at the pre-reform local office level; bars show 95% confidence intervals. The vertical lines separate the pre-reform, transition, and post-reform periods.

Figure 6: Heterogeneity by Sector



Notes: The figures report coefficients on the interaction between the reform-induced change in estimated appeal aggressiveness and the post-reform indicator, from specification (11), estimated separately by sector. Outcomes are indicated by the panel caption. The change is provincial minus local office aggressiveness, measured in units of 10 percentage points. The post-reform period is 2012–2019; the omitted period is 2005–2008, and the regression also includes an interaction for the 2009–2011 transition period. Regressions include firm and province-by-year fixed effects and municipality controls. Standard errors are clustered at the pre-reform local office level, and bars show 95% confidence intervals.

Table 1: Summary Statistics: Tax Offices of the Revenue Agency

	Mean	SD	Min	P25	Median	P75	Max	N
Panel A: Local Offices								
Cases	357.51	967.94	2.00	79.00	161.00	334.00	11,911.00	279
First-instance cases	300.22	812.46	2.00	67.00	135.00	287.00	10,205.00	279
First-instance cases won by firm (%)	26.73	12.24	5.63	18.64	24.07	33.33	100.00	279
First-instance cases won by office (%)	48.65	14.35	0.00	39.13	48.28	59.36	86.11	279
First-instance cases won by firm	81.37	223.59	2.00	15.00	36.00	69.00	2,438.00	279
Firm wins appealed by office (%)	61.10	19.42	0.00	50.00	61.54	74.24	100.00	279
Panel B: Provincial Offices								
Cases	1,415.41	2,391.05	50.00	524.00	732.00	1,504.00	19,114.00	105
First-instance cases	1,018.30	1,727.00	48.00	387.00	550.00	1,121.00	14,399.00	105
First-instance cases won by firm (%)	30.97	6.98	14.55	26.52	29.93	36.29	46.08	105
First-instance cases won by office (%)	38.22	8.51	17.76	32.50	38.01	43.90	64.23	105
First-instance cases won by firm	324.68	540.89	13.00	101.00	166.00	344.00	3,983.00	105
Firm wins appealed by office (%)	68.48	12.40	16.67	60.83	71.31	76.56	93.60	105

Notes: The table reports summary statistics across tax offices observed in the litigation data from 2005 to 2016. Panel A covers Local Offices and Panel B covers Provincial Offices. Case counts are totals over the observed period for each office. N is the number of offices. Six Local Offices have no cases observed in years 2005-2009 and are excluded from this table. The Aosta Provincial Office has no cases observed in years 2010-2016 and is excluded from this table.

Table 2: Summary Statistics: Firms in Main Sample

	Mean	SD	Min	P25	Median	P75	Max	N
Exit	0.06	0.23	0.00	0.00	0.00	0.00	1.00	8,901,772
Revenues	1,581.55	4,040.40	0.00	73.33	310.00	1,081.00	26,041.00	8,901,772
Employees	7.06	14.90	0.00	0.00	2.00	6.92	91.50	8,824,820
ROA	3.54	24.34	-100.00	0.00	3.40	9.68	79.70	8,901,772
1st-instance cases (x1000)	16.64	235.82	0.00	0.00	0.00	0.00	296,000.00	6,964,823
1st-instance cases won (x1000)	5.13	128.70	0.00	0.00	0.00	0.00	186,000.00	6,964,823
Cases won & appealed by office (x1000)	3.07	82.86	0.00	0.00	0.00	0.00	53,000.00	6,964,823
Provincial offices								83
Local offices								255

Notes: The table reports firm-year summary statistics for the main sample from 2005 to 2019. The sample includes firms assigned to local offices for which both Local and Provincial aggressiveness measures are available and where there are multiple Local Offices in the province at baseline. The sample covers 83 provincial offices and 255 local offices. Observations after exit are excluded, while the exit year is retained. Exit is an indicator for the year in which a firm exits. Revenues, employees, and return on assets (ROA) are reported in levels and are winsorized at the 1st and 99th percentiles, as in the main analysis. Revenues are measured in 1,000 of Euros. The litigation variables count first-instance cases per firm-year, cases won by the firm, and cases won by the firm and appealed by the office. Each count is multiplied by 1,000 for readability. Cases concern local or provincial offices of the Italian Revenue Agency and are assigned to the year of the outcome. Litigation statistics cover 2005–2016. N is the number of nonmissing firm-year observations for each variable.

Table 3: Estimated Effects of Aggressiveness on Firm Outcomes

	(1) Share Exited	(2) Log Revenues	(3) Log Number of Employees	(4) ROA
Δ Aggressiveness $\times$ Post	0.001	-0.012*	0.002	-0.103*
	(0.001)	(0.007)	(0.002)	(0.060)
Δ Aggressiveness $\times$ Transition	0.000	0.002	0.003**	0.036
	(0.001)	(0.005)	(0.002)	(0.043)
Observations	14,255,366	9,165,018	9,078,984	9,165,018
Pre-Period Outcome Mean	0.234	1389	5.980	2.721
R-squared	0.731	0.593	0.808	0.458
Province-Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Municipality Controls	Yes	Yes	Yes	Yes

Notes: This table reports firm-level estimates of specification (11). The change in aggressiveness is the empirical Bayes–adjusted appeal fixed effect of the post-reform provincial office minus that of the pre-reform local office, expressed in 10-percentage-point units. The coefficients interact this change with indicators for the transition period (2009–2011) and the post-reform period (2012–2019); the omitted period is 2005–2008. All specifications include firm and province-by-year fixed effects and the municipality controls described in Section 5.2. The outcome in column (1) is an indicator for having exited. The sample in columns (2) to (4) is restricted to active firms. These three outcomes are winsorized at the 1st and 99th percentiles. The reported pre-period outcome means are in levels. Revenues are measured in 1,000 of Euros. Standard errors, in parentheses, are clustered by pre-reform local office. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

APPENDIX

A.1 Additional Figures

Figure A1: Tax enforcement and litigation: data-generating process

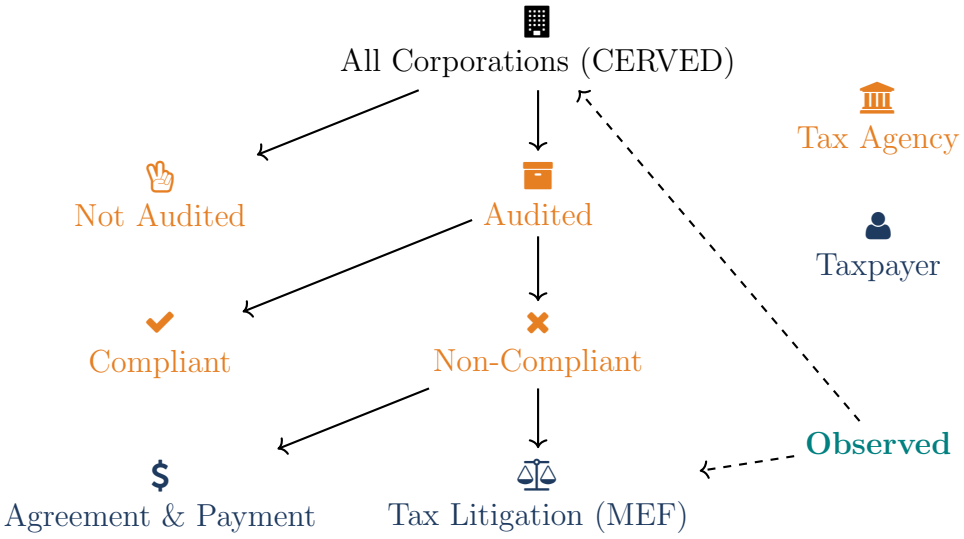
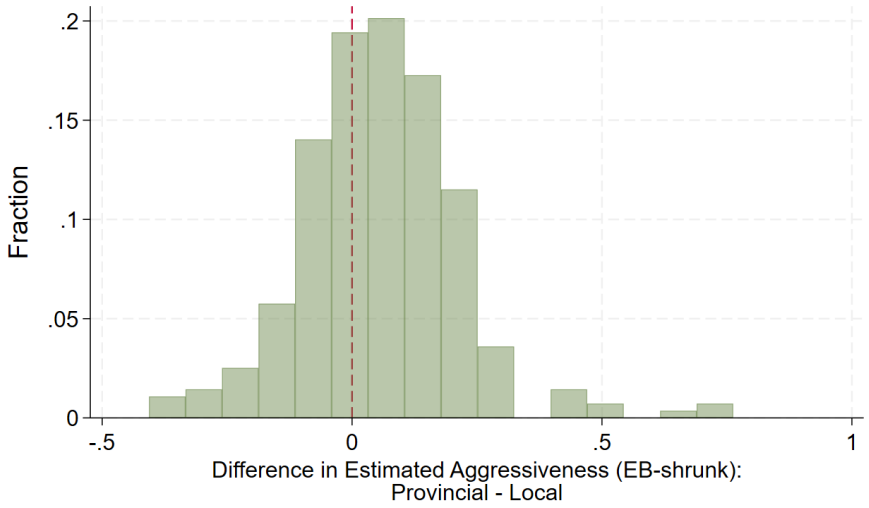


Figure A2: First-Instance Appeal Process



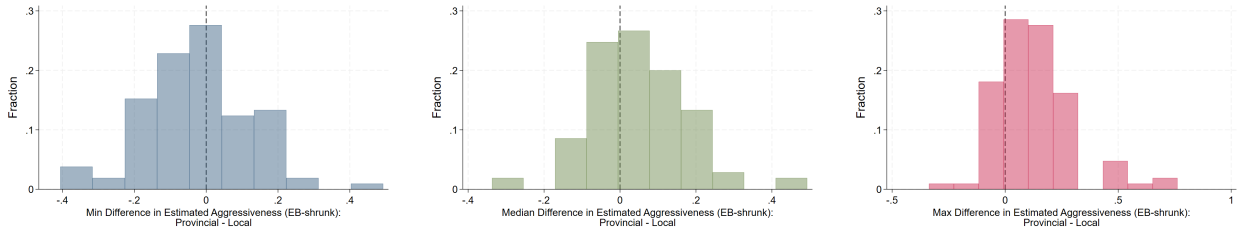
Figure A3: Local-to-Provincial Change in Aggressiveness



Notes: The figure shows the distribution of the difference between the estimated aggressiveness of the provincial office and that of the corresponding local office. Estimates are empirical Bayes–shrunk office effects. Each observation is a local–provincial office pair with a non-missing difference. Positive values indicate that the provincial office is more aggressive than the local office. The vertical line marks zero.

Figure A4: Local-to-Provincial Change in Aggressiveness: Statistics Across Provinces

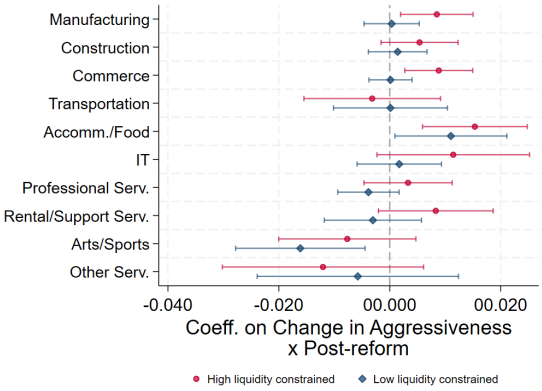
(a) Minimum Change (b) Median Change (c) Maximum Change



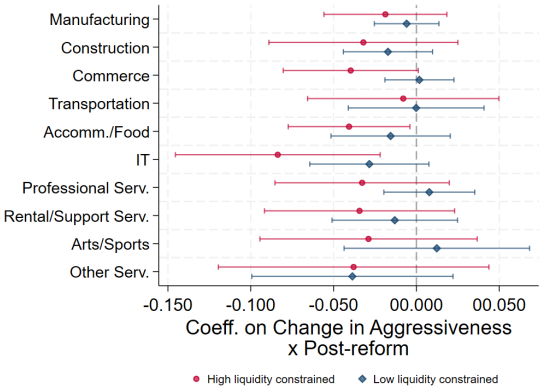
Notes: The figure shows the distribution across provinces of the minimum/median/maximum local-to-provincial change in estimated aggressiveness. For each province, the statistic is calculated across its observed local–provincial office pairs, with the change defined as provincial minus local aggressiveness. Each province contributes one observation. The vertical line marks zero.

Figure A5: Heterogeneity by Sector and Liquidity Constraints

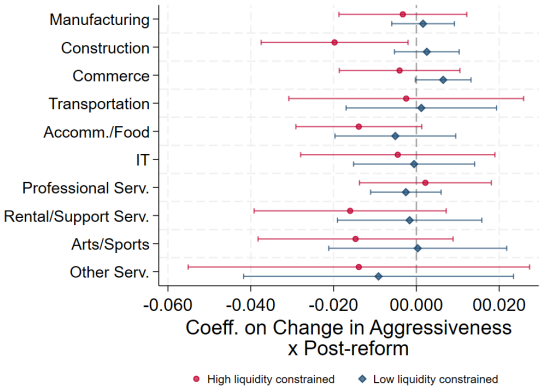
(a) Exit



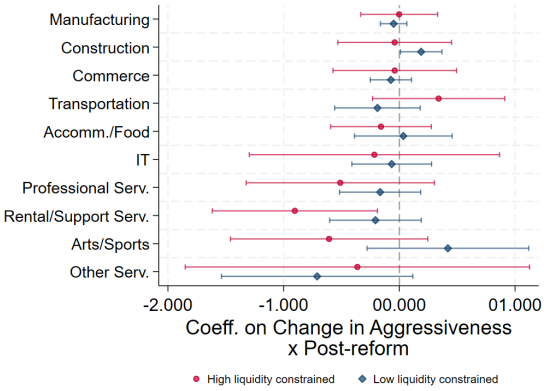
(b) Log(Revenues)



(c) Log(Number of Employees)



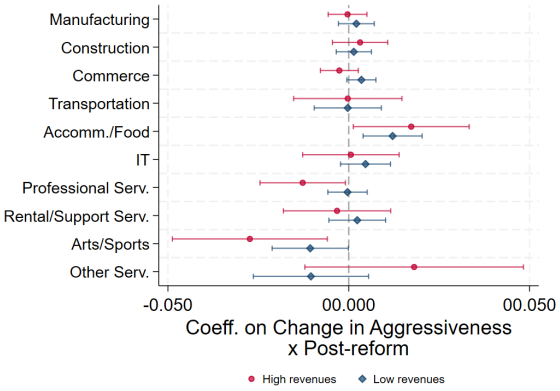
(d) ROA



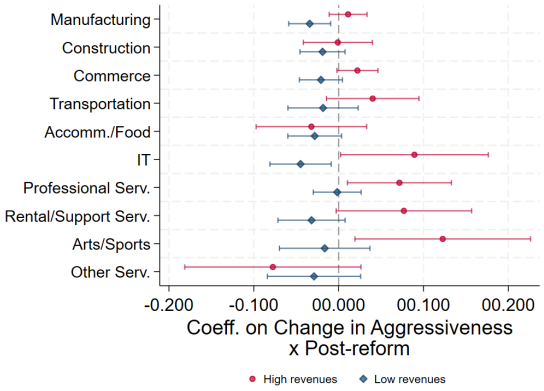
Notes: The figures report post-reform coefficients on the change in estimated appeal aggressiveness for firms with high and low pre-reform liquidity constraints, from specification (11), estimated jointly within each sector. Outcomes are indicated by the panel caption. The change is provincial minus local office aggressiveness, measured in units of 10 percentage points. The post-reform period is 2012–2019; the omitted period is 2005–2008, and the regression also includes an interaction for the 2009–2011 transition period. Regressions include firm and province-by-year fixed effects and municipality controls. Standard errors are clustered at the pre-reform local office level, and bars show 95% confidence intervals.

Figure A6: Heterogeneity by Sector and Firm Size (Pre-Reform Revenues)

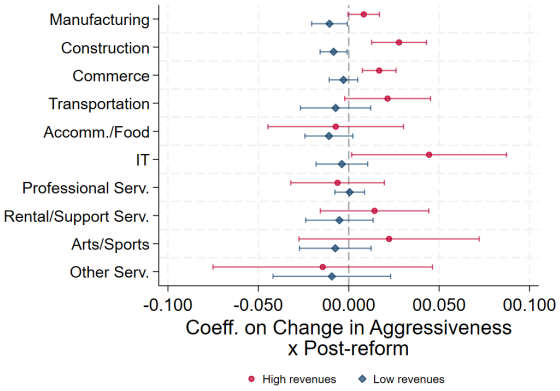
(a) Exit



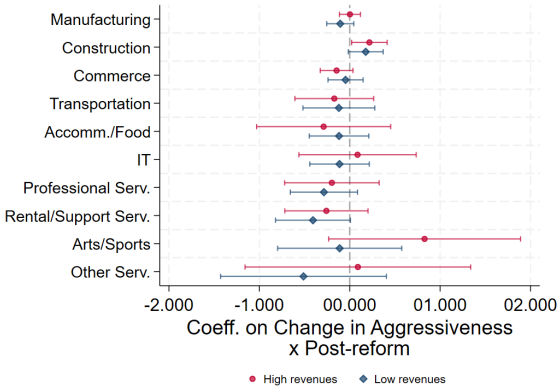
(b) Log(Revenues)



(c) Log(Number of Employees)



(d) ROA



Notes: The figures report post-reform coefficients on the change in estimated appeal aggressiveness for firms with high and low pre-reform revenues, from specification (11), estimated jointly within each sector. Outcomes are indicated by the panel caption. The change is provincial minus local office aggressiveness, measured in units of 10 percentage points. The post-reform period is 2012–2019; the omitted period is 2005–2008, and the regression also includes an interaction for the 2009–2011 transition period. Regressions include firm and province-by-year fixed effects and municipality controls. Standard errors are clustered at the pre-reform local office level, and bars show 95% confidence intervals.

A.2 Additional Tables

Table A1: Estimated Effects of Aggressiveness on Firm Outcomes by Liquidity Constraints

	(1) Share Exited	(2) Log Revenues	(3) Log Number of Employees	(4) ROA
Δ Aggressiveness × Post × High liquidity constrained	0.009***	-0.040***	-0.008**	-0.203
	(0.003)	(0.012)	(0.004)	(0.148)
Δ Aggressiveness × Post × Low liquidity constrained	0.001	-0.006	0.003	-0.056
	(0.002)	(0.007)	(0.002)	(0.059)
Observations	10,126,013	6,649,827	6,607,900	6,649,827
Pre-Period Outcome Mean	0.234	1389	5.980	2.721
R-squared	0.693	0.587	0.813	0.358
Province-Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Municipality Controls	Yes	Yes	Yes	Yes

Notes: This table reports firm-level estimates of specification (11), separately by pre-reform current liquidity. Firms are classified as highly liquidity constrained if their average current liquidity in 2005–2008 is below the 25th percentile; the remaining firms with non-missing average liquidity form the low-constraint group. The coefficients interact the reform-induced change in assigned tax-office aggressiveness with transition-period (2009–2011) and post-period (2012–2019) indicators separately for each group, including all double and triple interactions. The omitted period is 2005–2008. The change in aggressiveness is the empirical Bayes–adjusted appeal fixed effect of the post-reform provincial office minus that of the pre-reform local office, expressed in 10-percentage-point units. All specifications include firm and province-by-year fixed effects and the municipality controls described in Section 5.2. The outcome in column (1) is an indicator for having exited. The sample in columns (2) to (4) is restricted to active firms. These three outcomes are winsorized at the 1st and 99th percentiles. The reported pre-period outcome means are in levels. Revenues are measured in 1,000 of Euros. Standard errors, in parentheses, are clustered by pre-reform local office. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A2: Estimated Effects of Aggressiveness on Firm Outcomes by Firm Size (Pre-Reform Revenues)

	(1) Share Exited	(2) Log Revenues	(3) Log Number of Employees	(4) ROA
Δ Aggressiveness × Post × High revenues	-0.001	0.019	0.013***	-0.027
	(0.003)	(0.013)	(0.004)	(0.072)
Δ Aggressiveness × Post × Low revenues	0.003*	-0.032***	-0.006**	-0.130**
	(0.002)	(0.009)	(0.003)	(0.064)
Observations	10,134,399	6,653,531	6,611,473	6,653,531
Pre-Period Outcome Mean	0.234	1389	5.980	2.721
R-squared	0.694	0.589	0.814	0.354
Province-Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Municipality Controls	Yes	Yes	Yes	Yes

Notes: This table reports firm-level estimates of specification (11), separately by pre-reform revenues. High-revenue firms have average revenues in 2005–2008 above the 75th percentile; the remaining firms with non-missing average revenues form the low-revenue group. The coefficients interact the reform-induced change in assigned tax-office aggressiveness with transition-period (2009–2011) and post-period (2012–2019) indicators separately for each group, including all double and triple interactions. The omitted period is 2005–2008. The change in aggressiveness is the empirical Bayes–adjusted appeal fixed effect of the post-reform provincial office minus that of the pre-reform local office, expressed in 10-percentage-point units. All specifications include firm and province-by-year fixed effects and the municipality controls described in Section 5.2. These three outcomes are winsorized at the 1st and 99th percentiles. The reported pre-period outcome means are in levels. Revenues are measured in 1,000 of Euros. Standard errors, in parentheses, are clustered by pre-reform local office. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.